

ECSE 2500
Lee 17
March 27

Plan of next two classes

3/30 - Review class of HW 4-6

4/3 - Exam 2

Plan of today's class

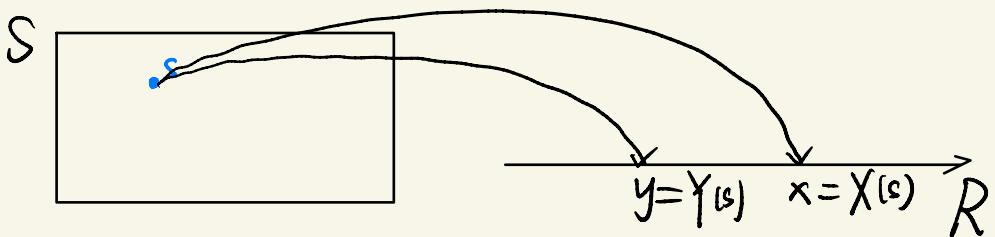
① CDF / PDF of two random variables

□ Two random variables

So far we've only looked at one RV at a time.

Now we will start talk about joint/two random variables.

That is, we do an experiment and obtain one outcome of the experiment which creates two real numbers from outcome



Example

One random experiment
generates two RVs

Flip coin n times. Define

$X(s) = \#$ of heads over n coin flipping

$Y(s) =$ position of the first head

Clearly, X is a Binomial random variable

Y is a Geometric random variable.

These two RVs follow different distributions, but there is a relationship between them. We will make this relationship clearly in the following note.

Example

We go to bank. Define

X is the length of time spent waiting in line

Y is the length of transactions

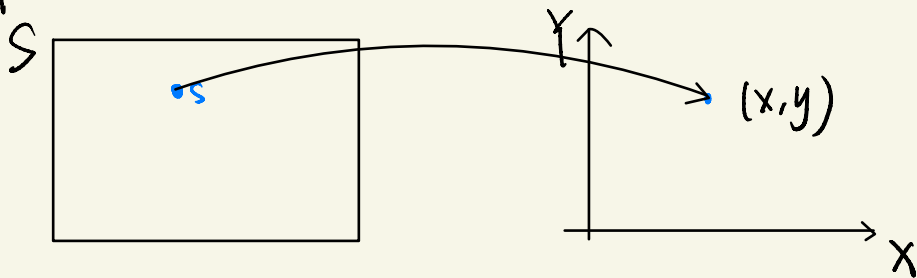
Example

Choose one person from a class. Define

X = height of the selected person

Y = weight of the selected person

In 2D, we can also think of (X, Y) providing a point in the plane:

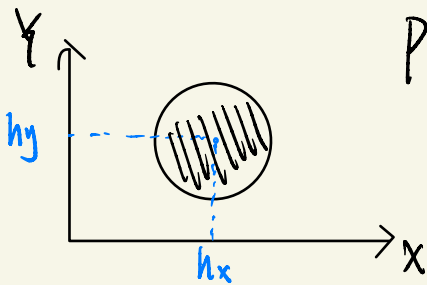


For two RVs, we will use almost the same terminology as on RV case: Events, CDF, PDF, PMF, Expectation, variance, etc.

- Joint Event (Extended from Event)

Example: In the case of using GPS, it will generate two RVs (X, Y) representing location information. What is

$P(\text{location is less than } \pm 5 \text{ km from origin})?$



$$\{(X, Y) \mid (X - h_x)^2 + (Y - h_y)^2 \leq 25\}$$

(h_x, h_y) location of origin

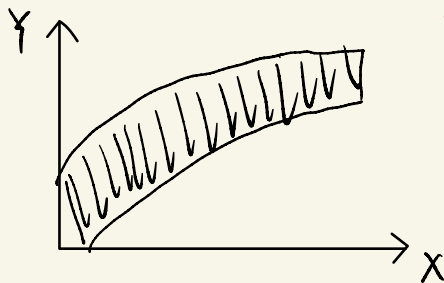
Example: In the height/weight example, what is

$P(\text{Body mass index of selected person is normal})?$

$$BMI = \frac{\text{Weight}}{(\text{height})^2}$$

$$\{(X, Y) \mid \frac{X}{Y^2} \in [n_1, n_2]\}$$

normal BMI range



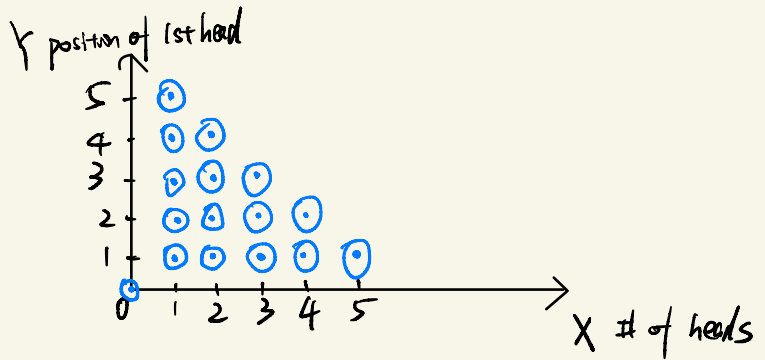
Example (Discrete)

In coin flip case, what is

$P(\text{less than 3 heads and first time head after 2 flips, } n=5)?$

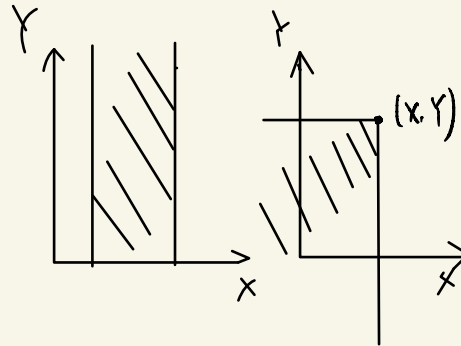
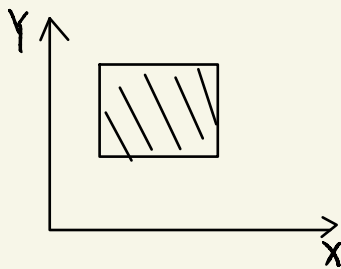
$$\{(X, Y) \mid X \leq 3, Y \geq 2\}$$

⊙ denotes the
outcome with non-zero
probability



In particular, we often interested in the "product"
from events of following form

$$E_X \cap E_Y$$



- Joint CDF of two random variables

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

$\swarrow$
big letter
 $\swarrow$
small letter

As before. CDF has the following properties

$$F_{X,Y}(\infty, \infty) = P(X \leq \infty, Y \leq \infty) = 1$$

$$F_{X,Y}(-\infty, y) = P(X \leq -\infty, Y \leq y) = 0$$

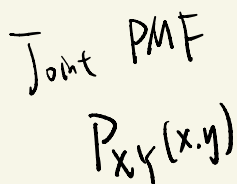
$$F_{X,Y}(x, -\infty) = P(X \leq x, Y \leq -\infty) = 0$$

- Joint PMF for two discrete RVs

$$P_{X,Y}(x,y) = P(X=x, Y=y)$$

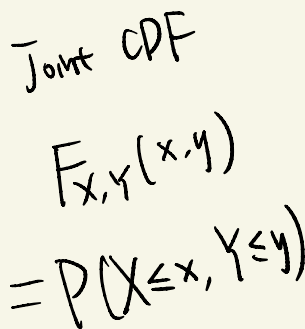
$\swarrow$
big case
 $\swarrow$
small case

In coin flip case, we have the following joint PMF function (assume a fair coin)



$$P(X=2, Y=4) = P(TTTHHH) = \frac{1}{32}$$

Then the joint CDF is what we get by summing up everything below and to the left of a given dot



$$F_{X,Y}(2,4) = P(X \leq 2, Y \leq 4)$$

$$= \sum_{\substack{x=0,1,2 \\ y=0,1,2,3,4}} P_{X,Y}(x,y) = \frac{1}{32} + \frac{1}{32} + \frac{1}{32} + \frac{1}{32} + \frac{1}{32}$$
$$\frac{1}{32} + \frac{1}{32} + \frac{1}{32} + \frac{1}{32} = \frac{5}{32}$$

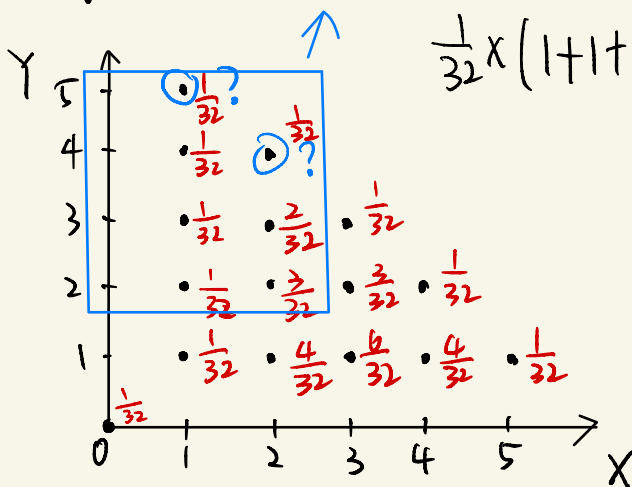
Q: Why we need joint PMF and CDF?

A: We can use this to compute the probability of any event.

For example, in our previous example, we want

$$P(X < 3 \text{ and } Y \geq 2) =$$

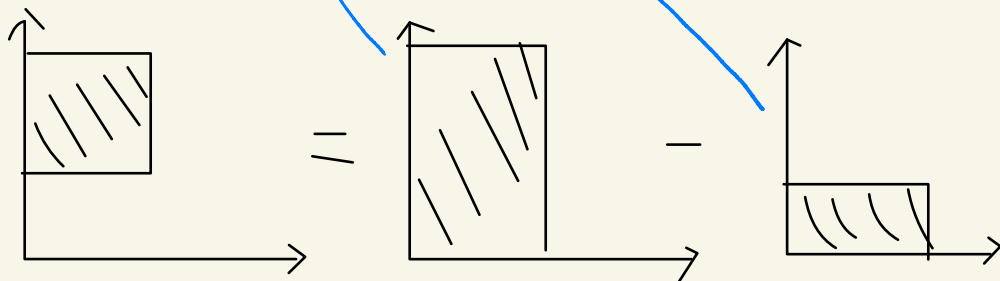
PMF
 $P_{X,Y}(x,y)$



$$\frac{1}{32} \times (1+1+1+1+1+2+5) = \frac{10}{32}$$

We can also compute this probability using joint CDF.

$$F_{X,Y}(2,5) - F_{X,Y}(2,1) = \frac{16}{32} - \frac{6}{32} = \frac{10}{32}$$



In general, we can compute the probability of a product-form event using CDF as follows

$$\begin{aligned}
 P(A) = & \quad \begin{array}{c} \text{Graph 1: A small rectangle in the first quadrant with corners } (a,b) \text{ and } (c,d). \text{ The area is shaded with diagonal lines and labeled } A. \end{array} \\
 & \quad = \begin{array}{c} \text{Graph 2: A large rectangle in the first quadrant with corners } (0,0) \text{ and } (c,d). \text{ The area is shaded with diagonal lines and labeled } F_{X,Y}(c,d). \end{array} \\
 & \quad - \begin{array}{c} \text{Graph 3: A rectangle in the first quadrant with corners } (0,0) \text{ and } (a,d). \text{ The area is shaded with diagonal lines and labeled } F_{X,Y}(a,d). \end{array} \\
 & \quad - \begin{array}{c} \text{Graph 4: A rectangle in the first quadrant with corners } (0,b) \text{ and } (c,b). \text{ The area is shaded with diagonal lines and labeled } F_{X,Y}(c,b). \end{array} \\
 & \quad + \begin{array}{c} \text{Graph 5: A small rectangle in the first quadrant with corners } (0,b) \text{ and } (a,b). \text{ The area is shaded with diagonal lines and labeled } F_{X,Y}(a,b). \end{array} \\
 & \quad = F_{X,Y}(c,d) - F_{X,Y}(a,d) - F_{X,Y}(c,b) + F_{X,Y}(a,b)
 \end{aligned}$$